From Anonymous Shapes to Named Places

A Tool for Braille and Place-Semantic Annotation of Tactile Maps

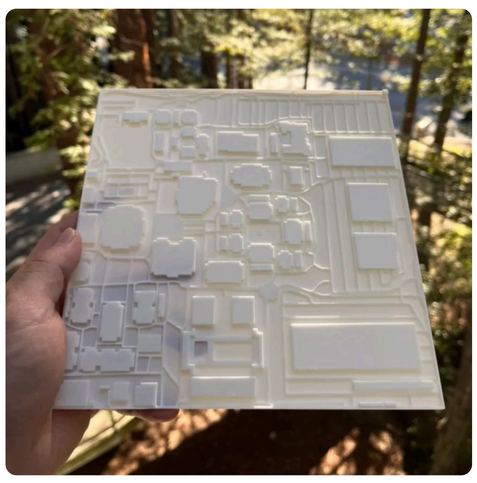
Li Liu[1], Ashmita Dua[1], Jiaming Qu[2,*], David T. Lee[1], Leilani H. Gilpin[1]
[1]University of California, Santa Cruz [2]Amazon ASSETS '26 · Demo 1266
Contact: lliu112@ucsc.edu

Accessible PDF

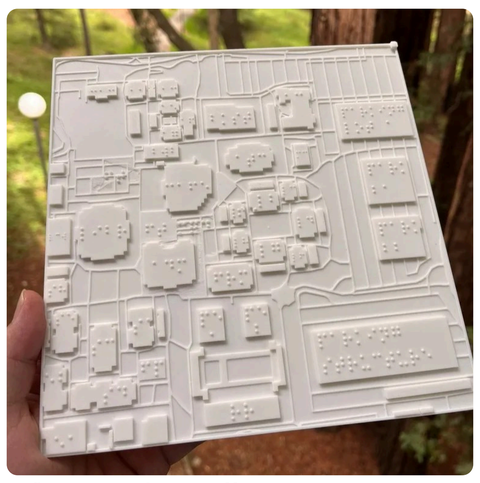
Try the tool

On a tactile map, a building under the finger is an anonymous shape.

Our tool adds Braille labels to tactile maps that already exist, so each shape on the print carries its own name.



Before · no labels



After · with Braille



Why we need it ## A tactile map gives the *where*. The *what* is hard to get.

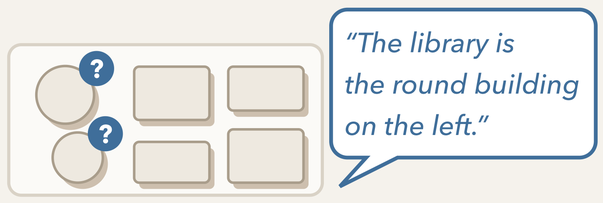
1



Felt, but not named

On a Touch Mapper† print, every building is a shape with no name.

2



Described, but hard to match

A spoken description is hard to tie to the right shape.

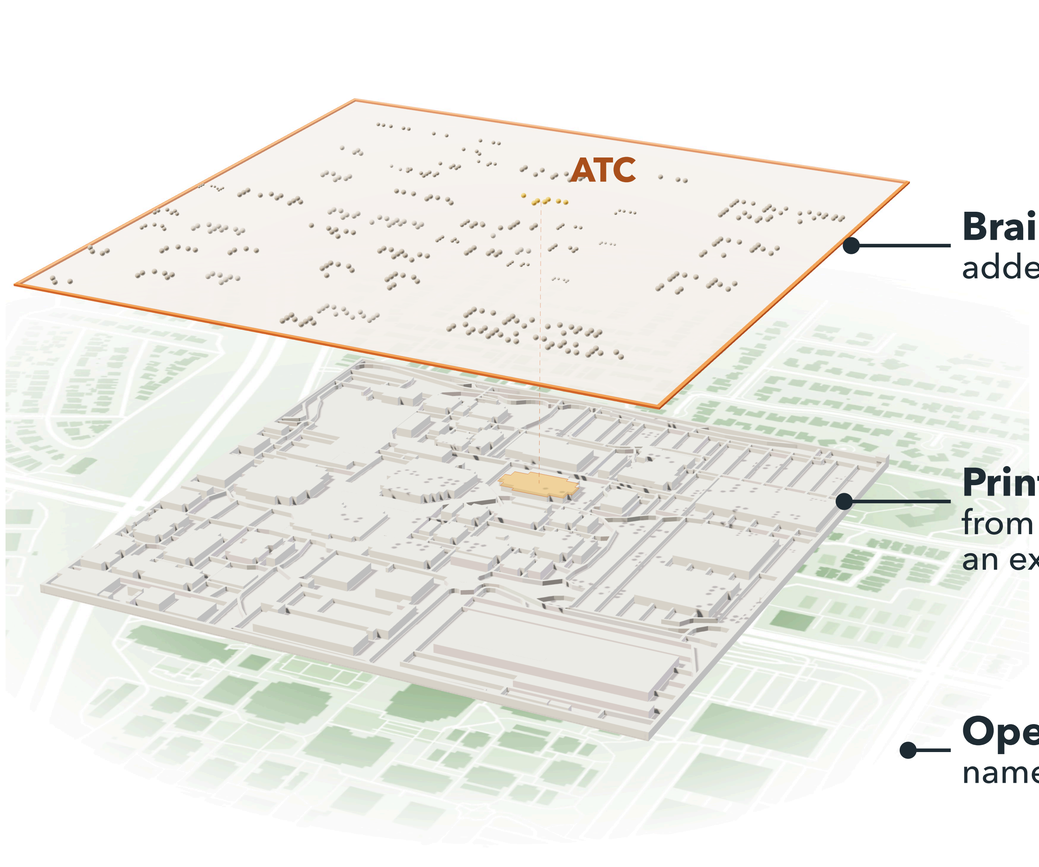
3



Hard to label by hand

Placing each label, printing it safely and editing the 3D model take time and skill.

What our tool does ## Braille becomes its own layer, linked to each place

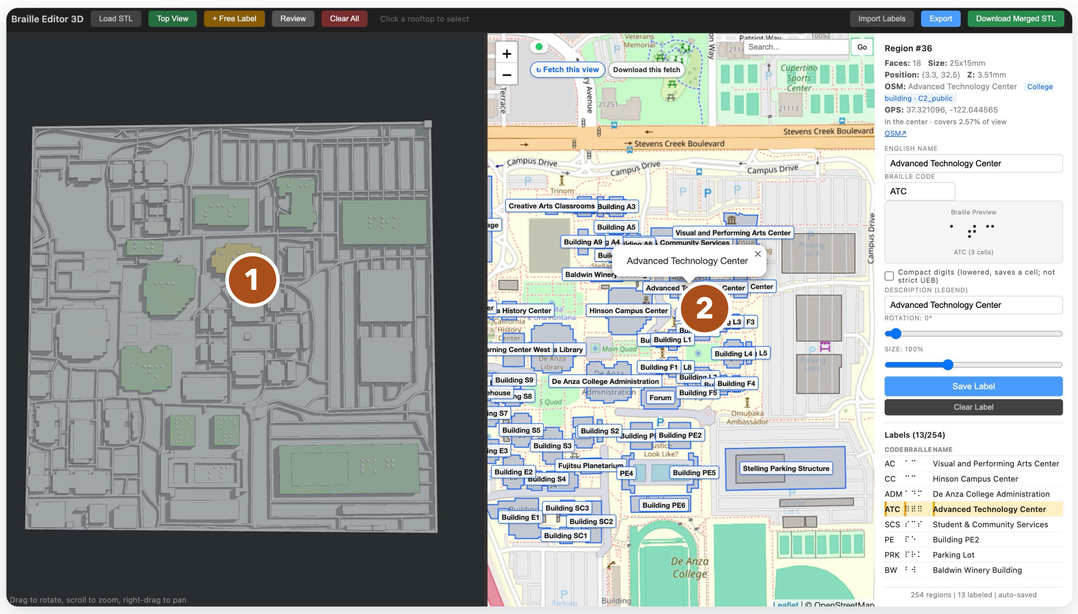


Braille layer
added with our tool

Printed base
from Touch Mapper, an existing tool

OpenStreetMap
names and places

3D model from Touch Mapper | **OpenStreetMap** | **Label**



1 Load the Touch Mapper model, click a building.

2 Click it on OpenStreetMap. Name and Braille fill in.

3 Review, then export. The dots are print-safe.

What it opens up ## One label, many ways to read it

Today

- Braille on the print
- Braille legend
- Print-ready 3D model

Each label links
a shape on the print
↓
a place on OpenStreetMap

Future

- Audio readout
- Symbols beyond Braille
- AI agents
- Shared Braille map

Project page leolee7.github.io/#project-tactile-annotation · **Paper** doi.org/10.1145/3797867.3841235

†Touch Mapper, an existing tool that makes tactile maps from OpenStreetMap: touch-mapper.org
*Work unrelated to position at Amazon.

© 2026 Copyright held by the owner/author(s). Map data © OpenStreetMap contributors.

UC SANTA CRUZ